LOGIC II

Week 5

LOGIC CIRCUIT

logic circuit an electronic circuit that has inputs and outputs that follow one of the Boolean operators.

When you write programs, you will often have to work out the correct **logic statements** for a given situation so that the right code runs.

Example:

RULES FROM THE PROBLEM

- Yellow leaves mainly near the soil → nitrogen
- Brown, small leaves → phosphorus
- Brown, normal-sized leaves → potassium

Start with advice set to 'nothing needed'. The relevant variables include colour and size.

ONE WAY TO EXPRESS THE RULES

```
SET advice TO 'nothing needed'
IF colour = 'yellow' THEN
    SET advice TO 'nitrogen'
ELSE
    IF colour = 'brown' AND size = 'small' THEN
        SET advice TO 'phosphorous'
    ELSE
        IF colour = 'brown' AND size = 'normal' THEN
            SET advice TO 'potassium'
        END IF
    END IF
END IF
```

Translate the words carefully: brown AND small means both conditions must be met.

Operator precedence: decide what happens first

1 BRACKETS	2 NOT	3 AND	4 OR
Evaluate the bracketed part first.	Reverse the Boolean value.	Both conditions must be true.	Either condition can make the result true.

Read a logic statement in context

Activity 20: who receives the 'Superstar!' message?

gender = 'female' AND (subject = 'computer science' OR subject = 'physics')

CASE	FEMALE?	CS OR PHYSICS?	MESSAGE?
Girl studying French	1	0	0
Girl studying computer science	1	1	1
Boy studying physics	0	1	0

WITHOUT THE BRACKETS

AND is evaluated before OR. So a boy studying physics would make the unbracketed expression TRUE — a different result.

Guided practice: revision-class condition

Activity 17 uses year group, grade and target to decide who is invited.

IF year = 11 AND (grade < target OR target > 7) THEN revision_class = true

Let Y = (year = 11), G = (grade < target), and T = (target > 7). Complete G OR T, then the final output.

Y	G	T	G OR T	OUTPUT
0	0	0	?	?
0	0	1	?	?
0	1	0	?	?
0	1	1	?	?
1	0	0	?	?
1	0	1	?	?
1	1	0	?	?
1	1	1	?	?

Explain the condition in words. Extension: write a program that asks for year, grade and target, then prints whether the student is invited.

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